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CLINICAL RESEARCH ARTICLE OPEN

# Dose-dependent effects of Alveofact with and without steroids on neutrophil extracellular traps in neonatal respiratory distress syndrome

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**BACKGROUND:** Neutrophil extracellular traps (NETs), peptidyl arginine deiminase 4 (PAD4), and oxidative stress play central roles in neonatal respiratory distress syndrome (RDS). This study investigated whether exogenous surfactant (Alveofact) combined with intratracheal budesonide modulates these pathways.
**METHODS:** Sixty preterm neonates with RDS were randomized into four groups to receive Alveofact 50 mg/kg or 100 mg/kg, with or without budesonide 0.25 mg/kg. Tracheal aspirates (TA) and blood samples were obtained at baseline and 24 h post-treatment, and an additional blood sample was obtained at 48 hours. NETs were measured by ELISA, oxidative stress by malondialdehyde (MDA) assay, and PAD4 expression by qPCR. Clinical outcomes, including hospital stay and mortality, were recorded.
**RESULTS:** The combination of Alveofact 100 mg/kg with budesonide significantly reduced NETs and secretory Immunoglobin A (sIgA) in TA, plasma NETs, MDA, and PAD4 expression compared with other treatment groups ($p < 0.05$). Plasma and TA NETs levels were strongly correlated across time points. No significant differences were observed in hospital stay, mortality, or complications among groups.
**CONCLUSION:** A combination of 100 mg/kg Alveofact with budesonide effectively attenuates neutrophil-driven inflammation and oxidative stress in neonatal RDS. The study was registered on the Clinical Trial Registration Site (ID: NCT06367881, first submitted 2024-04-07).
**CLINICAL TRIAL REGISTRATION:** https://www.clinicaltrials.gov/study/NCT06367881?term=NCT06367881&rank=1

*Pediatric Research*; https://doi.org/10.1038/s41390-026-04824-9

**IMPACT:**

- High-dose Alveofact + budesonide lowers NETs, PAD4, and oxidative stress.
- Plasma NETs strongly correlate with normalized tracheal aspirates NETs.
- First trial to test surfactant–steroid effects on neutrophil pathways in RDS.
- Therapy reduces neutrophil-mediated inflammation in preterm neonates.
- Plasma NETs may serve as a minimally invasive biomarker for RDS monitoring.

## INTRODUCTION

Neonatal respiratory distress syndrome (RDS) remains one of the leading causes of morbidity and mortality among premature infants worldwide, especially in regions with limited neonatal care and inconsistent antenatal steroid use.[1,2] The prevalence of RDS varies globally,[3] with higher incidences reported in developing countries, including Egypt rates.[4–6]

Current evidence indicates that RDS affects approximately 60-80% of infants born before 28 weeks of gestation,[7] with mortality rates remaining substantial despite therapeutic advances.[8,9] Contemporary management strategies for RDS encompass a multi-faceted approach, including antenatal corticosteroids, early continuous positive airway pressure (CPAP), and selective surfactant replacement therapy.[10,11] Alveofact (Bovactant), a natural bovine surfactant preparation containing phospholipids and surfactant proteins SP-B and SP-C, has demonstrated efficacy in reducing surface tension and improving pulmonary mechanics in preterm infants with RDS.[12,13] Alveofact significantly improves oxygenation and respiratory outcomes when administered early in the disease.[14–16] Combining surfactant therapy with corticosteroids enhances anti-inflammatory efficacy and improves pulmonary outcomes,[17] with emerging evidence suggesting enhanced anti-inflammatory effects.[18,19]

The role of neutrophil extracellular traps (NETs) is a web-like structure of DNA and histone complexes released by activated neutrophils that play a dual role in host defense and inflammatory

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Received: 27 August 2025 Revised: 14 December 2025 Accepted: 5 January 2026
Published online: 23 March 2026

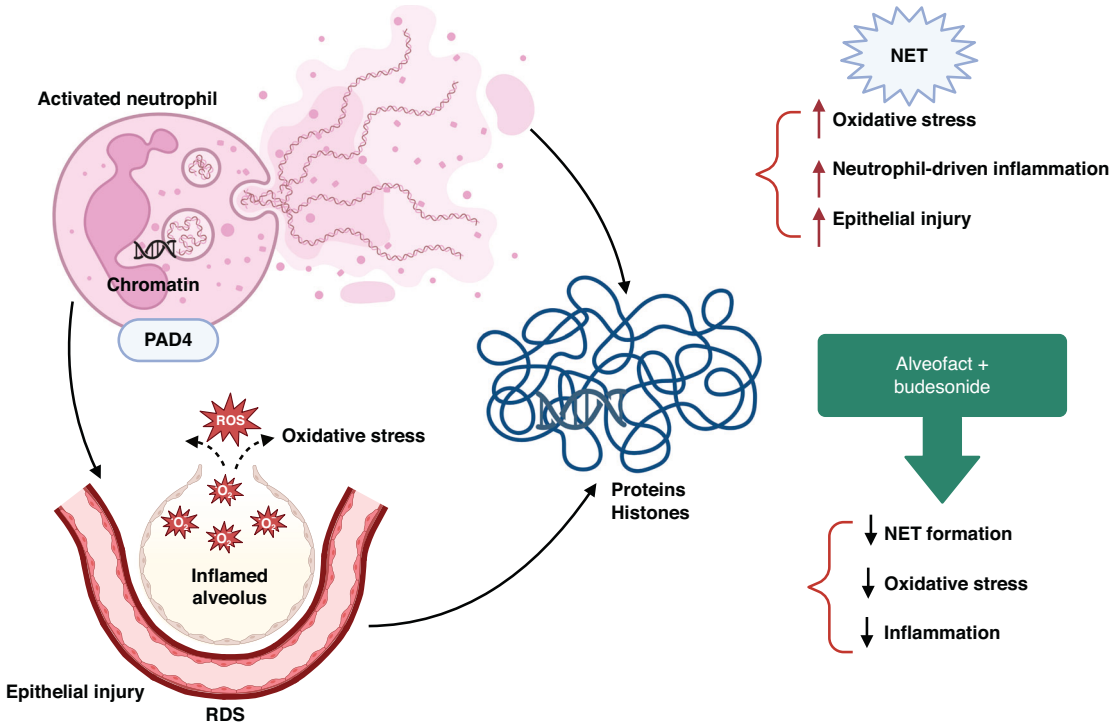


**Fig. 1 Proposed mechanism of NETs and PAD4 activation in Neonatal RDS.** Illustrative diagram showing the mechanistic role of neutrophil extracellular traps (NETs) and PAD4 activation in neonatal respiratory distress syndrome (RDS), and the modulatory effects of Alveofact ± budesonide In RDS, inflammatory stimuli in the immature lung trigger neutrophil recruitment and activation, leading to the release of NETs via PAD4-mediated chromatin decondensation. Excessive NET formation contributes to alveolar epithelial damage, surfactant inactivation, oxidative stress (via lipid peroxidation), and impaired gas exchange. High-dose Alveofact restores surfactant function and alveolar stability, while intratracheal budesonide exerts potent local anti-inflammatory effects, downregulating PAD4 expression and reducing NET formation. This combined regimen attenuates oxidative stress, preserves alveolar integrity, and facilitates improved respiratory function, potentially leading to better clinical recovery in preterm neonates with RDS.

injury. In RDS pathogenesis, it has emerged as a critical area of investigation, revealing complex inflammatory mechanisms underlying this condition.[20,21] NETs, consisting of decondensed chromatin decorated with cytotoxic proteins including neutrophil elastase and myeloperoxidase, are released by activated neutrophils as part of the innate immune response.[22] In the context of acute lung injury and RDS, excessive NET formation contributes to alveolar damage, increased vascular permeability, and persistent inflammation.[23,24] Studies have demonstrated elevated levels of NET-specific biomarkers in bronchoalveolar lavage fluid of patients with acute respiratory distress syndrome,[25] correlating with disease severity and clinical outcomes.[26] Importantly, natural surfactant preparations, including Alveofact and Curosurf, have been shown to exert dose-dependent inhibitory effects on NET formation,[25] suggesting an additional mechanism through which surfactant therapy may confer protective benefits beyond surface tension reduction.[27]

The involvement of activated neutrophils and peptidyl arginine deiminase 4 (PAD4) in RDS pathogenesis represents a novel therapeutic target.[28] PAD4, highly expressed in neutrophils, promotes histone citrullination and chromatin decondensation essential for NET formation.[29–31] The proposed mechanistic role of neutrophil extracellular traps (NETs) in neonatal respiratory distress syndrome (RDS) and the modulatory effect of Alveofact + budesonide therapy are presented in Fig. 1.

Despite advances in understanding RDS pathophysiology and treatment modalities, significant knowledge gaps remain regarding the optimal dosing strategies for surfactant therapy and the precise immunomodulatory mechanisms of Alveofact with or without corticosteroid co-administration.[32–34] The dose-dependent effects of Alveofact on immune cell function, particularly neutrophil activation and NET formation,[34] require systematic investigation to optimize therapeutic protocols.[35] Furthermore, subsequent NET formation remains largely unexplored. Understanding these mechanisms is crucial for developing evidence-based treatment strategies that maximize therapeutic benefits while minimizing potential adverse effects. This research gap necessitates comprehensive studies to elucidate the immunomodulatory properties of Alveofact across different dosing regimens and assess the synergistic potential of combination therapy with local steroids in the management of neonatal RDS.

The present study aims to evaluate the dose-dependent effects of Alveofact, with or without intratracheal budesonide, on NET formation, PAD4 expression, and oxidative stress in preterm infants with RDS. By integrating TA and plasma biomarker analyses, the study seeks to determine whether modulation of NET and PAD4 activity reflects local and systemic inflammatory responses and whether plasma NETs can serve as noninvasive biomarkers of treatment efficacy. This mechanistic insight may guide optimization of surfactant dosing strategies and support combined surfactant–steroid therapy as a dual-action approach for improving respiratory and immunologic outcomes in neonatal RDS.

## METHODS

### Study design & recruitment

This was an exploratory, randomized, double-arm, prospective controlled clinical trial with parallel assignment and four study arms, conducted in the Neonatal Intensive Care Unit (NICU) of Ain Shams University Children's Hospital, Cairo, Egypt. The trial was prospectively registered on ClinicalTrials.gov (Identifier NCT06367881) and approved by the Faculty of Medicine, Ain Shams University Research Ethics Committee (FMASU REC;

approval FMASU MD 185/2022). Recruitment targeted preterm neonates (gestational age ≤ 35 weeks) admitted to the NICU who fulfilled the European RDS consensus[33] RDS was confirmed by (i) characteristic clinical signs (tachypnea, nasal flaring, grunting, retractions, and cyanosis with reduced air entry), (ii) need for intubation during stabilization or worsening distress despite CPAP ≥ 6 cm $H_2O$ with $FiO_2$ >0.30 to maintain target saturations, and (iii) chest radiographic features consistent with RDS. Exclusion criteria comprised chromosomal anomalies, congenital heart defects, hemodynamically significant patent ductus arteriosus, early-onset sepsis or bacterial infection, congenital pneumonia, intraventricular hemorrhage, and parental refusal. After informed consent, a total of 60 preterm neonates with RDS were randomized and assigned in a 1:1:1:1 ratio (parallel allocation) to one of four arms: low-dose intratracheal Alveofact (50 mg/kg) without budesonide; low-dose Alveofact with budesonide (0.25 mg/kg); high-dose Alveofact (100 mg/kg) without budesonide; or high-dose Alveofact with budesonide (0.25 mg/kg). Randomization was performed using a computer-generated random sequence prepared by an independent statistician not involved in patient management. Allocation concealment was ensured through sequentially numbered, opaque, sealed envelopes, which were opened immediately before treatment. The clinical care team administering therapy was blinded to group allocation, and all laboratory investigators analyzing biological samples were likewise blinded to treatment assignment to maintain objectivity and reduce potential bias. As per protocol, all study groups could receive additional surfactant doses as clinically indicated if $FiO_2$ remained > 0.4 six hours after the initial dose. Masking of the outcome assessor was maintained. Additionally, 10 healthy preterm neonates matched for gestational age and sex were enrolled as a negative control group for normalization in gene expression analyses.

All neonates received standard NICU care. Prospective follow-up included detailed antenatal/perinatal history, serial clinical examinations, anthropometry, hemodynamic monitoring, ventilator settings, and $FiO_2$ on invasive and noninvasive support, duration of NICU stay, and major morbidities, including but not limited to Bronchopulmonary dysplasia (BPD), pneumothorax, pulmonary hemorrhage, intraventricular hemorrhage (IVH), sepsis, and mortality at discharge. RDS severity was quantified by the Oxygen Saturation Index (OSI), calculated as OSI = $FiO_2$ × (mean airway pressure) × 100 / $SpO_2$, measured at baseline and serially during follow-up. The study's primary outcome was neutrophil extracellular trap (NET) formation at 48 h; secondary outcomes included alveolar vs blood NET comparisons, ventilator days, lipid peroxidation (MDA), $FiO_2$ requirements, and NICU length of stay.

## Collection of samples (blood and bronchoalveolar lavage)

Sampling was performed at predefined time points to assess both local (pulmonary) and systemic responses to treatment. Tracheal aspirate (TA) and peripheral blood samples were collected at the time of admission (before treatment) and 24 h after Alveofact administration. TA (≥1.25 mL per time point) was obtained using sterile isotonic saline instillation through the endotracheal tube, followed by gentle aspiration. Blood samples (1 mL) were drawn into EDTA tubes, immediately placed on ice, and transported to the laboratory for prompt processing within 30 minutes. A detailed method was provided in the *Supplementary protocols*.

## Isolation of neutrophil granulocytes by density gradient centrifugation

Neutrophils were isolated from peripheral blood and TA using a discontinuous density gradient,[36] Ficoll-Paque™ PLUS (Cat. No. 17-1440-03, Cytiva, formerly GE Healthcare). The procedure was conducted according to the manufacturer's instructions. Viability was confirmed by Trypan Blue exclusion, and neutrophils were resuspended for downstream NET ELISA and PAD4 qPCR analysis. A detailed method was provided in the *Supplementary protocols*.

## Imaging

All infants underwent imaging as part of standardized assessment: chest radiography for RDS grading, cardiac size, and pulmonary congestion; echocardiography (2D, M-mode, color Doppler) to evaluate for PDA or structural heart disease; and transcranial ultrasonography with Doppler at admission and after Alveofact therapy to detect complications (e.g., IVH) and estimate cerebral blood flow Detailed method was provided in the *Supplementary protocols*.

## Measurement of NETs and Immunoglobulin A

Secretory IgA (sIgA) and NET concentrations were quantified directly in TA samples to account for potential dilutional variability associated with saline instillation during sampling. Parallel measurements of plasma NETs and serum IgA were also performed to allow comparison between airway and systemic inflammatory responses. TA samples were centrifuged at 300 × *g* for 10 min at 4 °C, and the supernatant was collected for analysis. NETs were quantified in TA and plasma using a sandwich ELISA specific for NET (MPO–DNA complex) according to the manufacturer's protocol (Elabscience® Human Neutrophil Extracellular Traps ELISA Kit, Cat. No. E-EL-H2462; Elabscience Biotechnology Inc., Houston, TX). Secretory IgA (sIgA) concentrations were measured in TA samples using the Human sIgA(Secretory Immunoglobulin A) ELISA Kit (cat no: E-EL-H1275), Elabscience Biotechnology Inc., Houston, TX. The ELISA kit used for tracheal aspirate secretory IgA specifically detects the secretory component–associated form of IgA and, according to the manufacturer, shows no significant cross-reactivity with monomeric serum IgA, whereas the serum IgA measured by assay is selective for total circulating IgA using an automated chemiluminescent immunoassay (Architect System, Abbott Laboratories, Green Oaks, IL).[37]

To further account for variability in TA dilution between samples, NET concentrations in TA were additionally normalized to the corresponding TA- sIgA concentration for each individual patient and time point. Normalized NET values were calculated by expressing TA NET concentration relative to TA sIgA (NET/sIgA) and are reported as ng/ng-sIgA. This normalization was performed at admission and 24 h after treatment. The percentage change in normalized NET/sIgA between baseline and 24 h was subsequently calculated to assess dilution-adjusted changes in airway NET burden following treatment.

## Measurement of Malondialdehyde (MDA) by Colorimetric assay

Malondialdehyde (MDA) levels in plasma and TA were measured using the Malondialdehyde (MDA) Assay Kit (Cat. No. E-BC-K025-M, Elabscience®, Houston, TX) according to the manufacturer's instructions.[38] A Detailed method was provided in the *Supplementary protocols*.

## Measurement of PAD4 gene expression by quantitative real-time PCR

Neutrophils isolated from TA and peripheral blood were pelleted by centrifugation at 400 × *g* for 5 min at 4 °C and lysed for total RNA extraction using the RNeasy® Mini Kit (Qiagen, Hilden, Germany; Cat. No. 74104). RNA concentration and purity were assessed spectrophotometrically (A260/A280 ratio). Reverse transcription was performed using the QuantiTect® Reverse Transcription Kit (Qiagen, Cat. No. 205311) according to the manufacturer's protocol, including genomic DNA elimination, cDNA synthesis at 42 °C for 15 min, and enzyme inactivation at 95 °C for 3 min. Quantitative PCR was carried out with the QuantiTect® SYBR® Green PCR Kit (Qiagen; Cat. No. 204143) on a real-time thermal cycler under the following cycling conditions: 95 °C for 15 min (enzyme activation), followed by 40 cycles of 94 °C for 15 s, 60 °C for 30 s, and 72 °C for 30 s. A melt-curve analysis confirmed amplification specificity. The primer sequences for PAD4 were: forward 5′-GCACAACATGGACTTCTACGTGG-3′ and the reverse 5′CACGCTGTCTTGGAACACCACA-3′. Glyceraldehyde 3-phosphate dehydrogenase (GAPDH) served as the endogenous reference gene. Relative expression was calculated using the 2^−ΔΔCt method with baseline samples as calibrators. All reactions were performed in duplicate, and no-template and reverse-transcription-minus controls were included in each run.[39] A detailed method was provided in the *Supplementary protocols*.

## Statistical analysis

Sample size estimation was performed using G\Power software version 3.1.9.7 (Heinrich Heine University, Düsseldorf, Germany). Assuming a medium effect size (f = 0.25) for differences in neutrophil extracellular trap (NET) levels between the four treatment groups, a power (1 − β) of 0.80, and an α-error probability of 0.05 in a repeated-measures ANOVA design (two measurements: baseline and 24 h post-treatment), the required sample size was calculated to be 52 neonates. To compensate for potential dropouts or early deaths, the sample size was increased by 15%, resulting in a total enrollment target of 60 patients (15 per group). An additional 10 healthy preterm neonates matched for gestational age and sex were included as a negative control group for normalization in gene expression analyses.

All statistical analyses were conducted using GraphPad Prism version 9.0 (GraphPad Software, San Diego, CA). Continuous variables were assessed for normality using the Shapiro–Wilk test and are presented as mean ± standard deviation (SD) for normally distributed data or median (interquartile range, IQR) for non-normally distributed data. Categorical variables are presented as frequencies and percentages. Between-group comparisons for continuous variables were performed using one-way ANOVA with Tukey's post-hoc test (parametric data) or the Kruskal–Wallis test with Dunn's correction (non-parametric data). Within-group comparisons over time (baseline vs. 24 h) were analyzed using paired t-tests or Wilcoxon signed-rank tests, as appropriate. Categorical variables were compared using the chi-square test or Fisher's exact test. Correlation analyses were performed using Pearson's or Spearman's rank correlation coefficients, depending on data distribution. In addition, linear regression analyses were conducted to explore the relationship between the percentage reduction in Oxygenation Severity Index (OSI) at 24 h post-treatment and the corresponding percentage reductions in TA NETs and plasma NETs. Separate simple linear models were first fitted for each predictor, followed by a multiple-regression model including both variables simultaneously. A two-tailed $p$-value < 0.05 was considered statistically significant.

## RESULTS

### Demographic characteristics of the studied cohort

The study included sixty preterm neonates diagnosed with respiratory distress syndrome (RDS), randomized into four treatment groups according to Alveofact dosage and budesonide co-administration, in addition to ten healthy preterm neonates serving as negative controls. The baseline demographic and perinatal characteristics are summarized in Table 1.

The mean gestational age and birth weight were comparable across all groups ($p = 0.881$ and $p = 0.972$, respectively). Distribution of birth weight categories (ELBW, VLBW, LBW) and sex showed no statistically significant differences. Most neonates in all groups were delivered via cesarean section (86.7–100%) and were from singleton pregnancies. The prevalence of intrauterine growth restriction (IUGR) and maternal comorbidities, including diabetes mellitus, severe preeclampsia, and antepartum hemorrhage, was low and evenly distributed between groups. Premature rupture of membranes >18 h, maternal inflammatory diseases, and infections were rare.

Antenatal corticosteroid administration was common, predominantly as a complete single course, and magnesium sulfate use was similar across groups. The only statistically significant difference observed was in the need for Alveofact redosing ($p = 0.028$), with Group 3 (high dose, no budesonide) showing the lowest proportion requiring an additional dose.

### Clinical and supportive care characteristics in the neonates with RDS

Baseline OSI scores, duration of hospitalization, and length of supplemental oxygen therapy were not significantly different between groups ($p > 0.05$). The duration of mechanical ventilation showed a non-significant trend toward reduction in neonates who received high-dose Alveofact without budesonide. The proportion of neonates requiring CPAP/NIV or invasive ventilation was comparable across groups. Inotropic support was infrequently required, with no significant intergroup differences. The incidence of persistent pulmonary hypertension (PPHN), BPD, pneumothorax, and mortality was low and similar among groups. Data are presented in Table 2.

### Effects of Alveofact dose and budesonide on NET and IgA levels in neonatal RDS

Tracheal aspirate (TA) secretory IgA (sIgA) levels declined over the first 24 hours in a dose- and steroid-dependent manner, with the greatest reduction observed in neonates treated with high-dose Alveofact plus budesonide. Although absolute sIgA concentrations at admission and 24 hours did not differ significantly between groups, the percentage reduction in sIgA differed significantly ($p = 0.001$), indicating a graded response to increasing surfactant dose and adjunct budesonide therapy (Table 3).

At admission, TA NET concentrations were comparable among the four RDS groups ($p = 0.064$), and plasma NET levels also showed no significant intergroup differences ($p = 0.461$) (Table 3; Fig. 2a, b). After 24 h of treatment, TA NET concentrations decreased in all groups, with the largest reduction observed in the high-dose Alveofact plus budesonide group (32.3%, $p = 0.0001$). Plasma NETs demonstrated a similar pattern, with a 32.5% reduction in the same group ($p = 0.0001$) (Table 3; Fig. 2c).

To account for potential dilutional variability in TA sampling, TA NET concentrations were normalized to TA sIgA on an individual-sample basis (NET/sIgA). At admission, normalized TA NET/sIgA values did not differ significantly between groups, indicating comparable baseline airway inflammatory burden after adjustment for dilution (Table 3). Absolute normalized NET/sIgA values at 24 h also showed no significant intergroup differences.

In contrast, the percentage change in normalized TA NET/sIgA from admission to 24 hours differed significantly between treatment groups ($p = 0.0001$). The greatest reduction in normalized NET/sIgA was observed in neonates receiving high-dose Alveofact with budesonide, followed by high-dose Alveofact alone, while smaller changes were seen in the low-dose groups (Table 3). These findings indicate that the reduction in TA NETs persists after adjustment for dilution and reflects a true treatment-related modulation of airway inflammation.

Plasma NET levels were compared between neonates with RDS at baseline ($n = 60$) and healthy controls ($n = 10$). A highly significant elevation was observed, with NET levels being approximately two-fold higher in the RDS group compared to controls (Table 3), consistent with systemic neutrophil activation in neonatal RDS.

### Regression analysis linking TA and plasma NET decline to OSI improvement at 24 h

As presented in Table 4, the regression analyses revealed a robust positive correlation between reductions in NET levels and improvement in oxygenation after 24 h of treatment. Decreases in TA NETs were closely linked to reductions in OSI, indicating that suppression of local pulmonary NET formation is strongly associated with rapid respiratory recovery. Likewise, plasmaNETs decline correlated significantly with OSI improvement, underscoring its value as a potential systemic biomarker of inflammatory control and therapeutic response.

When both TA and plasma NETs reductions were analyzed together, they explained 93% of the variability in OSI improvement. Both remained independent predictors, with TA NETs exerting a slightly stronger effect. This suggests that NETs modulation at both alveolar and systemic levels contributes synergistically to early clinical improvement, highlighting NET suppression as a key mechanism underlying the observed therapeutic benefit.

### Correlation between plasma and TA NETs levels in neonatal RDS

To assess whether plasma NET concentrations could serve as a surrogate marker for alveolar NET formation, we compared NET levels in paired TA and plasma samples obtained at admission and 24 h after treatment across all RDS groups.

At admission, NET levels in TA were consistently higher than those in plasma across all treatment groups ($p < 0.05$), reflecting the predominant local neutrophil activation within the pulmonary compartment. Following treatment, both compartments demonstrated significant reductions in NET concentrations, with the largest declines observed in neonates who received high-dose Alveofact + budesonide.

**Table 1.** Demographic characteristics of the studied cohort

| Variables | Low dose, no budesonide | Low dose + budesonide | High dose, no budesonide | High dose + budesonide | *p*-value |
|---|---|---|---|---|---|
| Delivery mode Vaginal CS | 2 (13.3) 13 (86.7) | 1 (6.7) 14 (93.3) | 0 15 (100) | 1 (6.7) 14 (93.3) | >0.05 |
| Gestational age (weeks) mean ± SD | 30.0 ± 1.56 | 30.2 ± 1.66 | 30.8 ± 1.74 | 31.0 ± 1.31 | >0.05 |
| Birth weight (kg)mean ± SD | 1.29 ± 1.0.22 | 1.30 ± 0.24 | 1.39 ± 0.32 | 1.42 ± 0.29 | >0.05 |
| Sex Male Female | 8 (53.3) 7 (46.7) | 10 (66.7) 5 (33.3) | 10 (66.7) 5 (33.3) | 7 (46.7) 8 (53.3) | >0.05 |
| Redosing Not required Single dose Two doses | 5(33.3) 6 (40) 4 (26.7) | 8 (53.3) 5 (33.3) 2 (13.3) | 13 (86.7) 2 (13.3) 0 | 12 (80) 3 (20) 0 | 0.028* |
| Birth weight (kg) ELBW VLBW LBW | 2 (13.3) 10 (66.7) 3 (20) | 3 (20) 9 (60) 3 (20) | 4 (26.7) 8 (53.3) 3 (20) | 2 (13.3) 8 (53.3) 5 (33.3) | >0.05 |
| IUGR No Yes | 10 (66.7) 5 (33.3) | 13 (86.7) 2 (13.3) | 12 (80) 3 (20) | 13 (86.7) 2 (13.3) | >0.05 |
| Multiple gestations Single fetus More than one fetus | 13 (86.7) 2 (13.3) | 14 (93.3) 1 (6.7) | 9 (60) 6 (40) | 12 (80) 3 (20) | >0.05 |
| Maternal Diabetes No Yes | 12 (80) 3 (20) | 14 (93.3) 1 (6.7) | 13 (86.7) 2 (13.3) | 13 (86.7) 2 (13.3) | >0.05 |
| PRM > 18 h NoYes | 14 (93.3) 1 (6.7) | 12 (80) 3 (20) | 15 (100) 0 | 14 (93.3) 1 (6.7) | >0.05 |
| Severe preeclampsia No Yes | 11 (73.3) 4 (26.7) | 11 (73.3) 4 (26.7) | 11 (73.3) 4 (26.7) | 7 (46.7) 8 (53.3) | >0.05 |
| Antepartum Hemorrhage No Yes | 11 (73.3) 4 (26.7) | 14 (93.3) 1 (6.7) | 11 (73.3) 4 (6.7) | 12 (80) 3 (20) | >0.05 |
| Maternal inflammatory diseases No Yes | 13 (86.7) 2 (13.3) | 15 (100) 0 | 14 (93.3) 1 (6.7) | 15 (100) 0 | >0.05 |
| Maternal infection No Yes | 14 (93.3) 1 (6.7) | 9 (60) 6 (40) | 13 (86.7) 2 (13.3) | 13 (86.7) 2 (13.3) | >0.05 |
| Prenatal steroids Didn't take Incomplete course Single complete course More than one course | 2 (13.3) 2 (13.3) 9 (60) 2 (13.3) | 5 (33.3) 1 (6.7) 8 (53.3) 1 (6.7) | 3 (20) 1 (6.7) 8 (53.3) 3 (20) | 3 (20) 30(20) 8 (53.3) 1 (6.7) | >0.05 |
| Maternal administration of Mg sulphate No Yes | 14 (93.3) 1 (6.7) | 9 (60) 6 (40) | 11 (73.3) 4 (26.7) | 12 (80) 3 (20) | >0.05 |

*ELBW* Extremely LBW (ELBW: birth weight <1.0 Kg), *VLBW* Very LBW (VLBW: birth weight <1.5 Kg), *LBW* Low BW (LBW: birth weight <2.5 Kg), IUGR intrauterine growth retardation, *PRM* Premature rupture membrane. The data are presented as mean and standard deviation (SD) for the numerical data and the distribution (n%)of categorical data for each group, including Low dose, no budesonide; Low dose + budesonide; High dose, no budesonide; and High dose + budesonide. The *p*-values for the numerical data were obtained using a one-way ANOVA test, and the *p*-values for the categorical data obtained using the Chi-square test ($\chi^2$ test) to compare the distribution across the four groups, *P*-value > 0.05 considered non-significant differences between compared groups, *: statistically significant difference between groups ($p < 0.05$).

Correlation analysis using Spearman's rank test demonstrated no significant association between normalized TA NETs/sIgA and plasma NET concentrations at admission ($p = 0.59$) or at 24 hours ($p = 0.88$). In contrast, the fold change in normalized TA NETs/sIgA showed a strong positive correlation with the fold change in plasma NETs ($r = 0.763$, $p = 0.001$), indicating that treatment-related changes in airway NET burden parallel systemic NET modulation, despite the absence of cross-sectional correlations at individual time points.

### High-dose Alveofact with budesonide yields the greatest reduction in plasma MDA levels in neonatal RDS

At admission, plasma malondialdehyde (MDA) concentrations, a biomarker of oxidative stress, were comparable across all RDS groups ($p = 0.178$) (Table 5, Fig. 3a). Following 24 h of therapy, MDA levels declined significantly in all treatment arms ($p = 0.0001$), with the most pronounced decrease observed in neonates who received high-dose Alveofact + budesonide, showing a 32.5% reduction (Table 5, Fig. 3b). Neonates receiving high-dose Alveofact, either with budesonide or without budesonide, exhibited greater reductions compared to low-dose groups, and the addition of budesonide further amplified the antioxidant effect. These findings indicate that high-dose Alveofact, particularly when combined with local budesonide, exerts a potent oxidative stress, reducing the severity of neonatal RDS.

Plasma MDA levels were significantly elevated in neonates with RDS, showing a 2.5-fold increase compared with healthy controls ($p = 0.0001$) (Table 5).

### High-dose Alveofact with budesonide Achieves the greatest reduction in PAD4 gene expression in neonatal RDS

At admission, plasma peptidyl arginine deiminase 4 (PAD4) gene expression levels were comparable across all RDS groups ($p = 0.251$) (Table 6, Fig. 3c). After 24 h of treatment, PAD4 expression declined in all groups, with the largest reduction observed in neonates receiving high-dose Alveofact + budesonide, showing a 12.3% decrease (Table 6, Fig. 3d). High-dose regimens of Alveofact with or without budesonide produced greater suppression compared to low-dose regimens, and the addition of budesonide further enhanced the anti-inflammatory effect. These results support a dose-dependent and synergistic action of high-dose Alveofact with local budesonide in modulating PAD4 activity, a key driver of NET formation and neutrophil-mediated inflammation in neonatal RDS.

In addition, plasma PAD4 gene expression was significantly higher in neonates with RDS, showing approximately a 4.0-fold increase compared with healthy controls ($p = 0.0001$) (Table 6).

## DISCUSSION

Respiratory distress syndrome (RDS) remains a leading cause of morbidity and mortality among preterm neonates,[40] primarily due to surfactant deficiency, immature lung architecture, and associated inflammatory injury.[41] Although exogenous surfactant therapy is the cornerstone of treatment, persistent oxidative stress and neutrophil-driven inflammation often limit recovery and contribute to complications such as bronchopulmonary dysplasia

**Table 2.** Clinical characteristics, respiratory support, and cardiac support among preterm neonates with RDS receiving different doses of Alveofact with or without budesonide

| Parameter | Low dose, no budesonide | Low dose + budesonide | High dose, no budesonide | High dose + budesonide | *p*-value |
|---|---|---|---|---|---|
| OSI score | | | | | |
| At admission | 6.7 ± 1.8 | 6.6 ± 1.9 | 6.5 ± 1.7 | 6.6 ± 1.6 | 0.981 |
| 24 h post- treatment | 2.3 ± 1.7 | 2.2 ± 1.4 | 2.8 ± 0.9 | 1.2 ± 0.3 | 0.06 |
| % of reduction at 24 h | 68.4 ± 11.7 | 60.5 ± 11.3a/d | 68.14 ± 13.4 d | 77.9 ± 5.6a/b/c | 0.001 |
| 48 h post treatment | 1.8 ± 0.6 | 0.9 ± 0.1a | 1.3 ± 0.3 d | 0.6 ± 0.1a/c | 0.005 |
| % of reduction 48 h | 75.8 ± 10.5 | 83.4 ± 7.2a/d | 84.9 ± 6.3a | 88.5 ± 5.6a/b | 0.001 |
| Hospital stay (days) | 20.3 ± 5.8 | 19.9 ± 6.1 | 18.7 ± 5.6 | 19.1 ± 5.4 | 0.774 |
| $O_2$ therapy duration (days) | 12.1 ± 3.5 | 11.8 ± 3.6 | 11.0 ± 3.4 | 11.2 ± 3.3 | 0.688 |
| Mechanical ventilation (days) | 8.1 ± 2.9 | 7.8 ± 3.0 | 6.7 ± 2.8 | 7.0 ± 2.7 | 0.129 |
| CPAP/NIV use, *n* (%) | 12 (80.0) | 11 (73.3) | 11 (73.3) | 12 (80.0) | 0.923 |
| Invasive ventilation, *n* (%) | 10 (66.7) | 9 (60.0) | 8 (53.3) | 9 (60.0) | 0.875 |
| Inotropic support, *n* (%) | 4 (26.7) | 3 (20.0) | 2 (13.3) | 3 (20.0) | 0.820 |
| PPHN, *n* (%) | 2 (13.3) | 2 (13.3) | 1 (6.7) | 1 (6.7) | 0.855 |
| Pneumothorax, *n* (%) | 1 (6.7) | 1 (6.7) | 1 (6.7) | 1 (6.7) | 1.000 |
| BDP, n (%) | 2 (13.3) | 0 | 1 (6.7) | 0 | 0.951 |
| Mortality, *n* (%) | 2 (13.3) | 2 (13.3) | 1 (6.7) | 2 (13.3) | 0.926 |

*OSI* Oxygen saturation index, *PPHN* persistent pulmonary hypertension, *CPAP* Continuous Positive Airway Pressure, *NIV* Non-Invasive Ventilation, *BDP* Bronchopulmonary dysplasia. data are presented in number & percentage within each group for categorical parameters and as mean ± standard deviation for numerical data, *P*-value > 0.05 considered non-significant differences between compared groups. [a]: statistical significance compared to the Low-dose Alveofact, [b]: statistical significance compared to the Low-dose Alveofact + budesonide, [c]: statistical significance compared to the high-dose Alveofact, [d]: statistical significance compared to the high-dose Alveofact + budesonide.

(BPD).[42,43] Increasing evidence implicates neutrophil extracellular traps (NETs),[44] lipid peroxidation products such as malondialdehyde (MDA), and pro-inflammatory enzymes like peptidyl arginine deiminase 4 (PAD4) as central mediators of acute lung injury in neonates.[45,46] However, the relationship between surfactant dosing, adjunct corticosteroid use, and modulation of these pathways in acute RDS is not fully understood.[47]

The current study expands on this concept by demonstrating that both surfactant dose and adjunct budesonide influence these inflammatory pathways. As illustrated in Fig. 1, excessive activation of PAD4 in neutrophils promotes NET formation and subsequent oxidative and tissue injury. High-dose Alveofact (100 mg/kg) combined with budesonide produced the greatest suppression of NETs, sIgA and PAD4 expression, particularly in tracheal aspirates (TA), indicating more effective attenuation of local pulmonary inflammation. The enhanced reduction in NETs, sIgA and PAD4 expression observed with high-dose Alveofact combined with budesonide may be attributed to the Marangoni effect, in which surface tension gradients facilitate distal delivery of budesonide through alveolar spreading. Increased exogenous surfactant concentration likely augments this gradient, thereby improving anti-inflammatory drug distribution within the peripheral airways. This mechanism aligns with prior experimental data demonstrating surfactant-assisted steroid transport to the distal lung regions.[48]

The observed decline in TA secretory IgA (sIgA) over the first 24 h after treatment reflects washout and dilution effects of saline and exogenous surfactant, rather than an isolated defect in IgA production.[49] As a key component of airway mucosal defense, transient reductions in sIgA are consistent with dilution and modulation of mucosal inflammation, particularly in the presence of higher surfactant doses and budesonide, which are known to attenuate pulmonary inflammation, indicating a potent local anti-inflammatory and immunomodulatory effect.[50,51] In contrast, serum IgA was assessed as an exploratory systemic marker, reflecting indirect immune modulation and inflammatory signaling, rather than direct mucosal immune response, especially when TA samples are unavailable.

In this context, evaluating airway NET concentrations after accounting for dilutional effects provides an important complementary perspective on the biological significance of the observed changes.[49] Importantly, normalization of tracheal aspirate NET concentrations to sIgA on an individual-sample basis demonstrated that the treatment-related reduction in airway NET burden persisted after adjustment for dilutional variability. While absolute NET concentrations declined across groups, the greatest reduction in normalized NET/sIgA occurred in neonates receiving high-dose surfactant combined with budesonide, indicating that the observed NET modulation reflects a true biological treatment effect rather than a sampling artifact. These findings further support the conclusion that combined high-dose surfactant and local corticosteroid therapy exerts a meaningful anti-inflammatory effect within the neonatal airway.

Importantly, the positive correlation between TA and plasma NETs suggests that systemic measurements can serve as a noninvasive reflection of alveolar inflammatory activity, supporting the hypothesis that airway inflammation can contribute to systemic inflammatory signaling through the release of neutrophil extracellular traps and PAD4 into circulation. In ventilated neonates, proximal airway injury and epithelial disruption may promote translocation of NET components into the bloodstream, reflecting the continuum between local and systemic

**Table 3.** Comparative analysis of, serum IgA, NET concentrations in TA and plasma (ng/mL) at baseline and 24 h post-treatment in RDS groups

| Parameter | Low dose, no budesonide | Low dose + budesonide | High dose, no budesonide | High dose + budesonide | *p*-value |
|---|---|---|---|---|---|
| TA-sIgA at admission "baseline" and 24 h post-treatment | | | | | |
| At admission | 0.16 ± 0.09 | 0.22 ± 0.08 | 0.22 ± 0.11 | 0.24 ± 0.07 | 0.06 |
| After 24 hours | 0.13 ± 0.08 | 0.17 ± 0.07 | 0.16 ± 0.08 | 0.13 ± 0.05 | 0.28 |
| % of reduction | 18.5 ± 7.5 | 23.2 ± 7.3 [d] | 29.6 ± 10.2[a/d] | 44.5 ± 11.2[a/b/c] | 0.001 |
| Serum IgA at admission "baseline" and 24 hours post-treatment | | | | | |
| At admission | 18.12 ± 6.78 | 19.59 ± 3.49 | 23.50 ± 7.54[a/d] | 22.37 ± 2.93[c] | 0.05 |
| After 24 hours | 12.13 ± 3.75 | 12.59 ± 2.72 | 13.44 ± 4.50 | 10.22 ± 1.28 | 0.06 |
| % of reduction | 30.97 ± 9.90 | 35.74 ± 6.62[a/c/d] | 42.32 ± 5.57[a/b/d] | 52.5 ± 8.86[a/b/c] | 0.003 |
| TA NETs at admission "baseline" and 24 hours post-treatment | | | | | |
| At admission | 9.76 ± 2.10 | 9.28 ± 2.20 | 11.30 ± 2.60 | 11.10 ± 2.30 | 0.064 |
| After 24 hours | 9.14 ± 2.00 | 8.12 ± 2.10 | 8.64 ± 2.40 | 7.60 ± 2.00 | 0.259 |
| % of reduction | 9.14 ± 2.20 | 12.6 ± 4.8[a/c/d] | 23.5 ± 8.2[a/d] | **32.3 ± 9.6 [a/b/c]** | **0.0001** |
| Normalized TA NETs to sIgA at admission "baseline" and 24 hours post-treatment | | | | | |
| At admission (ng/ng-sIgA) | 39.67 ± 12.2 | 43.2 ± 11.9 | 43.9 ± 13.2 | 45.0 ± 14.6 | 0.706 |
| After 24 hours (ng/ng-sIgA) | 37.6 ± 10.8 | 39.6 ± 10.2 | 36.8 ± 12.1 | 33.4 ± 12.9 | 0.539 |
| % of change | 4.7 ± 2.1 | 8.1 ± 4.5[c] | 16.2 ± 7.2[a/b/d] | 26.8 ± 9.5[a/b/c] | **0.0001** |
| Plasma NETs at admission "baseline" and 24 hours post-treatment | | | | | |
| At admission | 10.50 ± 1.70 | 9.70 ± 1.70 | 10.10 ± 2.10 | 9.20 ± 2.50 | 0.461 |
| After 24 hours | 9.50 ± 1.50 | 8.21 ± 1.60[c/d] | 7.80 ± 1.88[a] | **7.20 ± 2.07[a]** | **0.006** |
| % of reduction | 9.2 ± 3.5 | 17.3 ± 6.3[a/c/d] | 22.9 ± 5.8[a/b/d] | **32.5 ± 5.3[a/c/d]** | **0.0001** |
| Plasma-NETs | **NRDS** | | **Healthy controls** | | |
| mean ± SD | 10.2 ± 2.02 | | 4.78 ± 2.03 | | **0.0001** |

Data are mean ± SD. Comparative analysis: Kruskal–Wallis test. $p < 0.05$ is considered significant. [a]: statistical significance compared to the Low-dose Alveofact, [b]: statistical significance compared to the Low-dose Alveofact + budesonide, [c]: statistical significance compared to the high-dose Alveofact, [d]: statistical significance compared to the high-dose Alveofact + budesonide. *Abbreviations:* TA: Tracheal aspirate, NETs: Neutrophil extracellular traps, sIgA: secretory IgA.

inflammation. These findings are consistent with earlier animal and human studies linking bronchial inflammation to plasma NET marker elevation.[52]

This mechanistic link supports prior reports that corticosteroids down-regulate PAD4-dependent NETosis and oxidative burst,[53–55] while surfactant reduces inflammatory mediator release by restoring alveolar stability.[56]

The marked reduction in MDA further confirms that high-dose surfactant plus budesonide mitigates lipid peroxidation, consistent with experimental data showing synergistic antioxidant effects when surfactant is combined with intratracheal corticosteroids. This is biologically plausible, as surfactants can stabilize alveolar membranes and reduce oxygen radical–induced injury, while corticosteroids can attenuate oxidative pathways via transcriptional regulation of antioxidant enzymes.[57–59] Our findings align with animal studies showing that combined surfactant and steroid therapy reduces oxidative stress markers more effectively than surfactant alone.[60]

PAD4 expression, a key driver of NET formation, was highest at baseline in RDS neonates and decreased significantly post-treatment, with the largest suppression observed in the high-dose Alveofact + budesonide group. This supports the hypothesis that corticosteroid co-administration directly interferes with NETosis pathways, likely via transcriptional suppression of PAD4. Previous research has demonstrated that PAD4 inhibition reduces lung injury in models of acute respiratory distress,[61,62] and our results extend this observation to the neonatal RDS setting.

While clinical outcomes such as hospital stay and mortality did not differ significantly between groups, the biochemical improvements observed within 24 hours indicate an early anti-inflammatory response that may translate into better long-term pulmonary recovery. This may reflect the limited sample size per group, reducing the study's statistical power for clinical endpoints. The absence of significant differences also suggests that short-term biochemical improvements do not necessarily translate into immediate survival or length-of-stay benefits within the early postnatal window. Similar trends have been reported in studies where early budesonide–surfactant therapy reduced subsequent BPD incidence.[63,64] Previous randomized trials using surfactant–budesonide mixtures in preterm infants have reported reductions in bronchopulmonary dysplasia incidence and improved extubation success rates.[65,66] Nevertheless, the trends observed merit further exploration in larger, multicenter cohorts.

Importantly, this is the first clinical investigation to quantify NETs simultaneously in TA and plasma of preterm neonates with RDS, linking these markers to PAD4 expression and oxidative stress. Demonstrating that plasma NETs reflect TA levels supports their potential as minimally invasive biomarkers of lung inflammation and treatment response.

This study has several limitations. First, the relatively small sample size in each treatment arm limited the statistical power for detecting differences in hospital stay, mortality, and complication rates, raising the possibility of type II error. Second, the single-center design may limit the generalizability to other neonatal intensive care units with differing clinical practices.

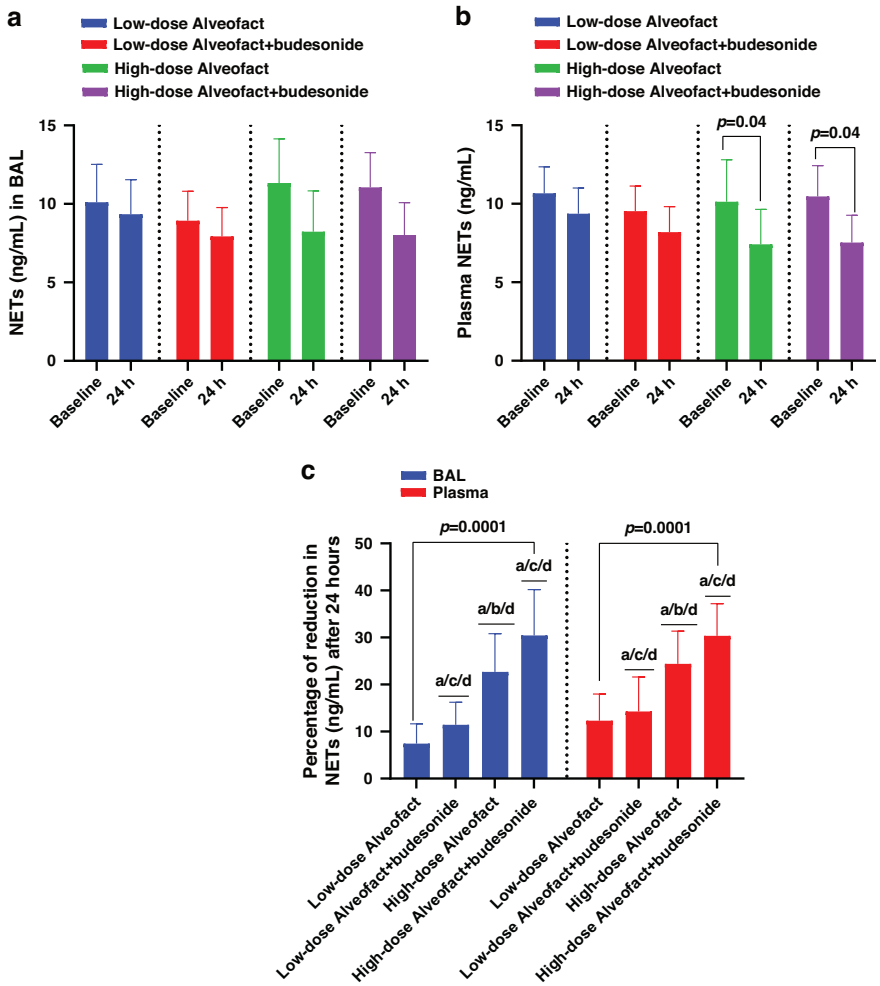


**Fig. 2 NET concentrations and reduction in Neonates with RDS following therapeutic regimens.** Bar charts showing neutrophil extracellular trap (NET) concentrations (pg/mL) in (**a**) Tracheal aspirate (TA) and (**b**) plasma of neonates with respiratory distress syndrome (RDS) at admission (before treatment) and 24 hours after intervention with four different therapeutic regimens. **c** Percentage reduction in NET levels at 24 hours relative to baseline for each treatment group. Data are presented as mean ± SD. Statistical significance was set at $p < 0.05$. Superscripts indicate significance compared with: [a] – low-dose Alveofact; [b] – low-dose Alveofact + budesonide; [c] – high-dose Alveofact; [d] – high-dose Alveofact + budesonide.

**Table 4.** Regression analyses examining the relationship between % reduction in OSI (24 h) and % reduction in NET levels

| Predictor | B | SE B | β | *p* | 95% CI for B |
|---|---|---|---|---|---|
| Model 1 – TA NETs (24 h) | | | | | |
| Constant | 44.27 | 2.81 | — | 0.001 | (38.66 - 49.88) |
| % Reduction in TA NETs (24 h) | 1.12 | 0.09 | 0.94 | 0.001 | (0.94 - 1.31) |
| Model 2 – Plasma NETs (24 h) | | | | | |
| Constant | 43.05 | 3.10 | — | 0.001 | (36.80 - 49.30) |
| % Reduction in Plasma-NETs (24 h) | 0.98 | 0.08 | 0.92 | 0.001 | (0.82 - 1.15) |
| Model 3 – Combined (TA + Plasma NETs) | | | | | |
| Constant | 41.91 | 2.92 | — | 0.001 | (36.05 - 47.77) |
| % Reduction in TA NETs (24 h) | 0.74 | 0.12 | 0.62 | 0.001 | (0.50 - 0.98) |
| % Reduction in Plasma NETs (24 h) | 0.42 | 0.10 | 0.41 | 0.001 | (0.22 - 0.62) |

*B* Unstandardized Coefficient, *β* Standardized Coefficient, *SE* Standard Error, *CI* Confidence Interval.

Third, although random allocation minimized bias, residual confounding from antenatal exposures, supportive care variations, and comorbidities cannot be excluded. Fourth, follow-up was limited to hospital discharge, precluding assessment of long-term respiratory or neurodevelopmental outcomes. Finally, while we examined NETs, MDA, and PAD4 expression, additional inflammatory and proteomic analyses could provide a more complete mechanistic understanding.

**Table 5.** Comparative analysis of plasma MDA concentrations (nmol/mL) at baseline and 24 h post-treatment in RDS groups

| Parameter | Low dose, no budesonide) | Low dose + budesonide | High dose, no budesonide | High dose + budesonide | *p*-value |
|---|---|---|---|---|---|
| At admission | 33.4 ± 7.25 | 35.3 ± 9.1 | 33.8 ± 7.4 | 29.58 ± 2.5 | 0.178 |
| After 24 hours | 28.2 ± 6.5 | 27.0 ± 5.6[c/d] | 24.5 ± 5.7[a/b/d] | **19.8 ± 3.4[a/b/c]** | **0.0001** |
| % Reduction | 15.7 ± 4.2 | 22.5 ± 5.9[a/c/d] | 27.8 ± 5.8[a/b/d] | **32.5 ± 7.5[a/b/c]** | **0.0001** |
| Plasma-NETs | **NRDS** | | **Healthy controls** | | |
| mean ± SD | 33.1 ± 7.48 | | 13.3 ± 3.79 | | **0.0001** |

Data are mean ± SD. Comparative analysis: Kruskal–Wallis test. $p < 0.05$ is considered significant. [a]: statistical significance compared to the Low-dose Alveofact, [b]: statistical significance compared to the Low-dose Alveofact + budesonide, [c]: statistical significance compared to the high-dose Alveofact, [d]: statistical significance compared to the high-dose Alveofact + budesonide. *TA* Tracheal aspirate, *MDA* Malonaldehyde.

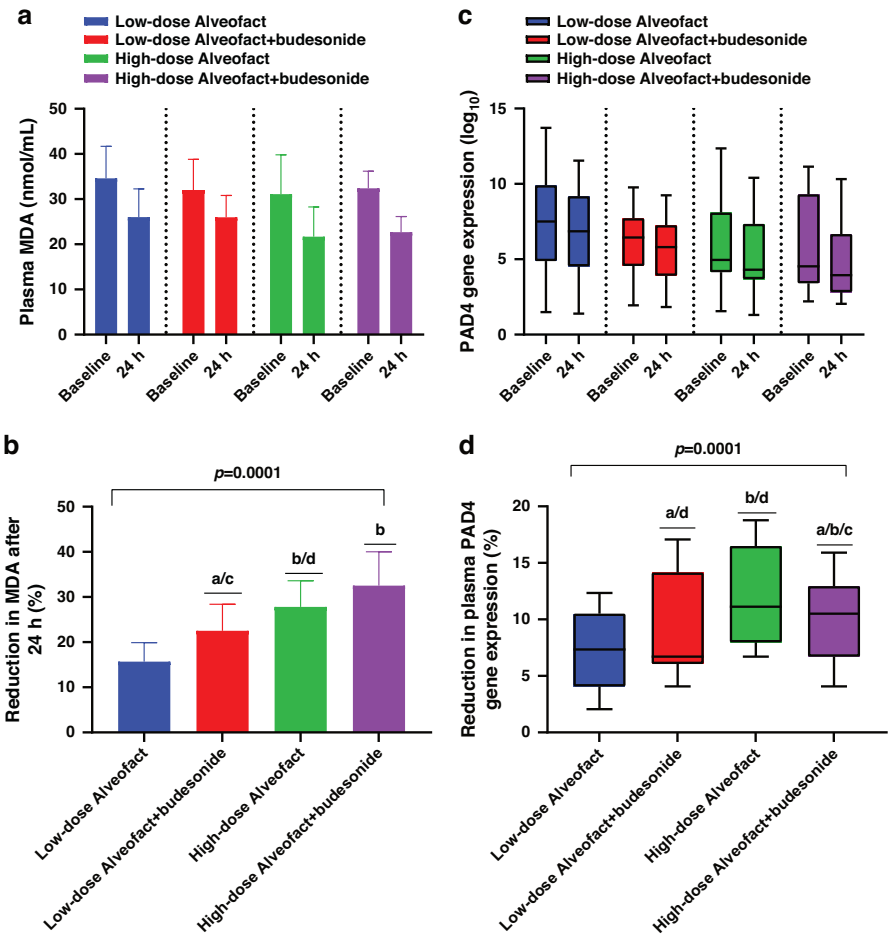


**Fig. 3 Effects of different therapeutic regimens on MDA concentrations and PAD4 gene expression in Neonates with RDS.** (**a**) Bar chart showing malondialdehyde (MDA) concentrations (nmol/mL) in neonates with respiratory distress syndrome (RDS) at admission (before treatment) and 24 hours after intervention with four different therapeutic regimens. Data are presented as mean ± SD. (**b**) Percentage reduction in MDA levels at 24 h relative to admission for each treatment group. (**c**) Boxplots depicting plasma peptidyl arginine deiminase 4 (PAD4) gene expression ($log_{10}$-transformed) in neonates with RDS at admission and 24 h after treatment with the same four regimens. Data are presented as median and range due to non-normal distribution. (**d**) Percentage reduction in PAD4 gene expression at 24 hours relative to baseline. Statistical significance was set at $p < 0.05$. Superscripts indicate significance compared with [a] – low-dose Alveofact; [b] – low-dose Alveofact + budesonide; [c] – high-dose Alveofact; [d] – high-dose Alveofact + budesonide.

## CONCLUSION

In neonates with RDS, high-dose Alveofact combined with budesonide resulted in the most pronounced reductions in NETs, MDA, and PAD4 expression, with plasma NET levels closely reflecting those in TA. Importantly, dilution-adjusted analysis using normalized TA NETs to sIgA confirmed that reductions in airway NET burden persisted after accounting for sampling variability, indicating a true biological treatment effect rather than an artifact of lavage dilution. While clinical outcomes did not differ significantly between treatment groups, the observed

**Table 6.** Comparative analysis of plasma PAD4 gene expression ($Log_{10}$) at baseline and 24 h post-treatment in RDS groups

| PAD4 ($Log_{10}$) | Low dose, no budesonide | Low dose + budesonide | High dose, no budesonide | High dose + budesonide | *p*-value |
|---|---|---|---|---|---|
| At admission | 7.87 (4.90) | 6.39 (2.95) | 4.75 (5.70) | 4.75 (4.19) | 0.251 |
| After 24 hours | 7.25 (4.62) | 5.45 (2.71) | 4.46 (5.50) | 3.93 (3.60) | 0.059 |
| % Reduction | 6.69 (1.29) | 9.87 (8.10)[a/c/d] | 7.34 (4.40)[b/d] | 12.3 (5.40)[a]/[b/c] | 0.0001* |
| Plasma NETs | **NRDS** | | **Healthy controls** | | |
| Median | 4.95 (2.77) | | 1.07(0.72) | | 0.001 |

Data are median (IQR). Comparative analysis: Mann–Whitney test; $p < 0.05$ significant. Data are mean ± SD. Comparative analysis: Kruskal–Wallis test. $p < 0.05$ is considered significant. [a]: statistical significance compared to the Low-dose Alveofact, [b]: statistical significance compared to the Low-dose Alveofact + budesonide, [c]: statistical significance compared to the high-dose Alveofact, [d]: statistical significance compared to the high-dose Alveofact + budesonide. *TA* Tracheal aspirate, *PAD4* peptidyl.

biomarker improvements suggest potential for long-term benefit, supporting further evaluation in larger trials. Plasma NET measurement may offer a less invasive alternative to TA for monitoring pulmonary inflammation in this population.

## DATA AVAILABILITY

The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

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## AUTHOR CONTRIBUTIONS

Conceptualization, NE, and HA; Formal analysis, GIG; Investigation, AMR, and DR; Methodology, AMR, and NE; Project administration, GIG; Writing – original draft, NE; Writing – review & editing, AMR, NE, HA, GIG, and DR.

## FUNDING

Open access funding provided by The Science, Technology & Innovation Funding Authority (STDF) in cooperation with The Egyptian Knowledge Bank (EKB).

## COMPETING INTERESTS

The authors declare no competing interests.

## ETHICS APPROVAL AND CONSENT FOR PARTICIPATION

The study was conducted in accordance with the Declaration of Helsinki and its later amendments (revised in October 2018). Approval to conduct this study was given by the Research Ethics Committee at the Faculty of Medicine, Ain Shams University, Egypt, Approval FMASU MD 185/2022. The clinical trial was prospectively registered on www.ClinicalTrials.gov (ID: NCT06367881). Informed consent was obtained from all participants prior to their involvement in the study.

## ADDITIONAL INFORMATION

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